

TECHNIQUE MASTERING EXERCISES

B - INEQUALITIES

1.

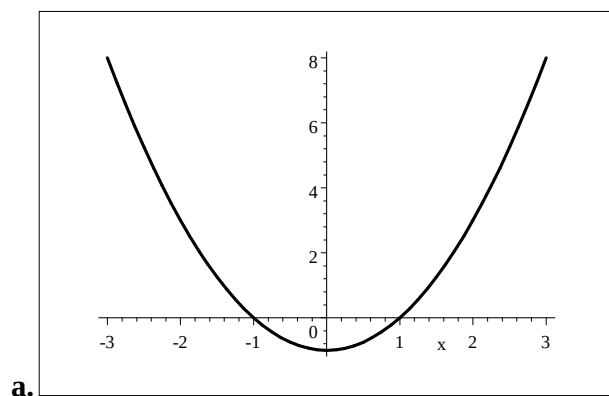
a. $x \in (-\infty, 3]$ b. $x \in (-\infty, 3)$ c. $x \in (-\infty, -3) \cup [3, \infty)$

d. $x \in (-\infty, -\frac{1}{3}) \cup (0, \infty)$ e. $x \in (3, \infty)$ f. $x \in [3, \infty)$

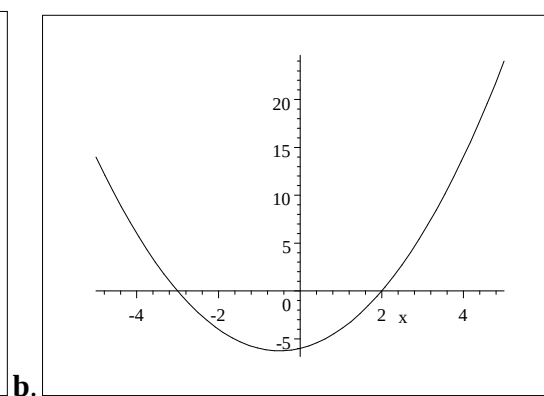
2. Use graphs to solve the inequalities and write your answers in interval notation.

a. $x \in (-\infty, -1) \cup (1, \infty)$ b. $x \in (-\infty, -3) \cup (2, \infty)$

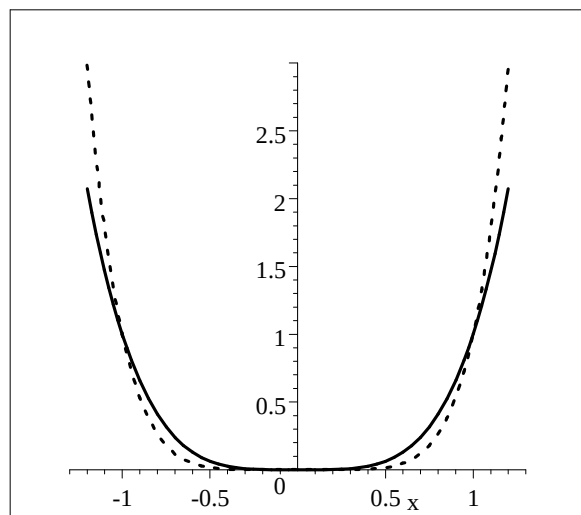
c. $x \in (-\infty, -1] \cup \{0\} \cup [1, \infty)$ d. $x \in (-1, 0) \cup (1, \infty)$



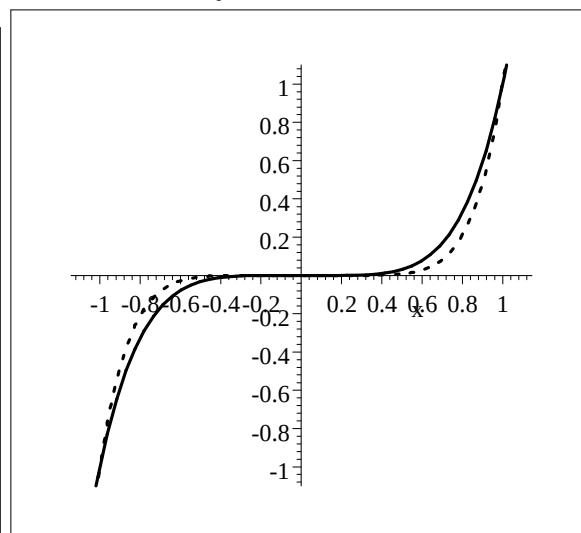
$$y = x^2 - 1$$



$$y = x^2 + x - 6$$



$$y = x^6 \text{ is dotted}$$



$$y = x^7 \text{ is dotted}$$